
Curriculum Vitae

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Languages:	Norwegian (B2); English (fluent); Greek (native); Korean (native)	Linkedin:	linkedin.com/in/pavloslee95

Personal Summary

I am a Post-Doctoral Fellow specializing in autonomous systems control using artificial intelligence (AI) and risk assessment techniques. My research topics span the areas of autonomous vehicles & robotics; safe & trustworthy autonomy; motion planning & data-driven control; machine learning; deep reinforcement learning (DRL); online risk modeling & risk awareness.

Education

2020-2024	PhD	Department of Naval Architecture, Ocean, and Marine Engineering, UK University of Strathclyde · Research topic: Safe e-navigation and collision avoidance of autonomous ships · Supervisors: Prof. Gerasimos Theotokatos; Prof. Evangelos Boulougouris
2013-2019	MEng	School of Naval Architecture and Marine Engineering, National Technical University of Athens, Greece · Research topic: Bioinspired microbubble lubrication system for the reduction of drag resistance of ships · Supervisor: Prof. George S. Triantafyllou · Admission: 3rd place for 2013 class

Working Experience

2024-	Postdoctoral Fellow	Department of Marine Technology, Norwegian University of Science and Technology, Norway · Project: SFI AutoShip (No. 309230/NOK240m), Research Council of Norway · Job description: Research on risk-aware AI agents with supervisory control for safety-critical operations of autonomous systems; Lecturer and Teaching Assistant; Design and correction of course assignments and exams; Research dissemination · Supervisors: Prof. Ingrid Bouwer Utne; Assoc. Prof. Ekaterina Kim
2020-2022	Research Assistant	Department of Naval Architecture, Ocean, and Marine Engineering, UK University of Strathclyde · Project: AUTOSHIP (No. 815012/€29.5m), Horizon 2020 · Job description: Risk assessment and development of a training framework for shore control center operators and onboard crews in autonomous ship operations; Deliverables
2017	Trainee Superintendent Engineer	Technical Department of Newbuildings, Maran Gas Maritime Inc., Greece Daewoo Heavy Industries liaison · Job description: Support the monitoring and verification of technical specifications with shipyards
2016-2023	Interpreter	International Exhibitions & Events: Posidonia; European Conference and Exhibition on Optical Communication; Embassies, Greece & UK · Job description: Interpreter for various none-profit and business sectors
2016-2019	Teacher	Athens Korean School, Greece · Job description: Curriculum design, teaching Korean language courses for various proficiency levels, and event organizing; Exam Invigilator; Treasurer

Academic Teaching

2026 Fall	NTNU	Risk Analysis and Safety Management of Marine Systems (TMR4130/MSc) · Lecturer on AI for risk assessment and Bayesian belief networks for risk modeling; Teaching Assistant	Norway
2025 Fall	NTNU	Risk Analysis and Safety Management of Marine Systems (TMR4130/MSc) · Lecturer on AI for risk assessment; Teaching Assistant for 43 MSc students & PhD candidates	Norway
2021 Fall	UoS	Marine Engineering with Specialization in Autonomous Marine Vehicles (NM844/MSc) · Lecturer on guidance, navigation, and control of autonomous ships	UK

Industry Collaboration

2025-	DNV	Development of assurance frameworks to build trust in black-box AI agents through adaptive testing, risk-awareness, and self-explainability	Norway
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Grants & External Funding

2026	KSSEA	Development of funding proposal for the Korean Federation of Science and Technology Societies to foster bilateral Scandinavian-Korean R&D initiatives involving joint research projects and scientific events (NOK270k)	Scandinavia & Korea
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Scientific Dissemination & Outreach

Selected list:

2025	Invited Lecturer	KIRD, Strategic Training of Researchers in Naval Architecture and Ocean Engineering for Overseas Studies (<i>online</i>) · Topic: Cyber-physical systems, digital twin, and DRL for shipping 4.0	Korea
2026	Invited Speaker	2 nd KONNEKT e-Workshop on AI4MI (Korea-Nordic e-Workshop for Networking & Ties) (<i>online</i>) · Topic: AI in autonomous shipping – A DRL perspective	The Nordics & Korea
2026	Presentation	Europe-Korea Conference 2026 · Topic: Towards risk-aware AI agents for safety-critical operations of autonomous ships	France
2025	Invited Speaker	Ocean Technology Day & AUR-Lab Day · Is your AI agent risk aware?	Norway
2025	Invited Speaker	KSSEA · Deep reinforcement learning & autonomous ships	Norway
2022	Presentation	The International Maritime and Port Technology and Development Conference and the 4th International Conference on Maritime Autonomous Surface Ships (<i>online</i>) Topic: Path following of autonomous ships using deep reinforcement learning	Singapore

Professional Networks

2025-	Organizing Committee	International Workshop for Autonomous System Safety (IWASS)	International
2025-	Organizing Committee	Korean–Scandinavian Scientists and Engineers Association (KSSEA)	Scandinavia & Korea
2020-2020-2024	Member	Europe–Korea Marine and Ocean Engineers Association (EKMOA)	EU & Korea
	Member	Korean Scientists and Engineers Association in the United Kingdom (KSEAUK)	UK & Korea
2013-	Member	Hellenic Institute of Marine Technology (HIMT)	Greece
2013-	Member	Society of Naval Architects and Marine Engineers (SNAME) – The Greek Section	Greece

Volunteering

2026	Session Chair	Europe–Korea Marine and Ocean Engineers Association Science & Technology (Session MO6), AI-Driven Future of Science and Technology, The Europe–Korea Conference on Science and Technology 2026	France
2026	Organizing Committee	The 7 th International Workshop for Autonomous System Safety 2026	Portugal
2026	Session Chair	Risk, Reliability and Safety of Autonomous Systems (Special Session 18), The 36 th European Safety and Reliability Conference	Portugal
2026	Peer Reviewer	The 36 th European Safety and Reliability Conference: 5 reviews	Portugal
2025-	Organizing Committee	Semi-annual Academic Meetings & Events, The Korean–Scandinavian Scientists and Engineers Association	Norway
2025	Session Chair	Autonomous Systems III (Session 16L), The 35 th European Safety and Reliability Conference & The 33 rd Society for Risk Analysis Europe Conference	Norway
2025	Organizing Committee	Safety, Reliability, and Security of Autonomous Systems Panel (Session 10L), The 35 th European Safety and Reliability Conference & The 33 rd Society for Risk Analysis Europe Conference	Norway
2024	Organizer Assistant	The 1 st International Workshop on Maritime and Shipping Industries Contemporary Challenges and Future Prospects – Decarbonization and Digitalization	Greece
2022-	Peer Reviewer	Ocean Engineering: 4 reviews; Journal of Marine Engineering and Technology: 1 review; Marine Geodesy: 1 review	-
2022-2023	Team Leader	Autonomous Control Team, Marine Robotics Society	UK
2021	Organizer Assistant	The 1 st International Conference on the Stability and Safety of Ships and Ocean Vehicles	UK

Awards & Certificates

2024	Shortlisted	Young Researcher Competition, TRA Visions · Topic: Robust reactive collision avoidance system for autonomous ships amidst limited sensory information and unclassified surface obstacles	Dublin
2023	Team Spirit Award	The Autonomous Ship Challenge, Njord · Description: Design and operation of autonomous surface vessels for navigation and docking challenges	Norway
2022	2nd Prize	Young Researcher Competition, TRA Visions (€3k) · Topic: Trustworthy autonomous collision avoidance decision-making system for maritime autonomous surface ships Sponsor: Waterborne Technology Platform	Portugal
2020-2023	Studentship Award	University Research Excellence Award (£46k) · Description: Projects with research excellence that support the University's strategic research ambitions	UK
2020	Certificate	Safety at Sea Workshop, Hellenic Marine Environment Protection Association · Description: Enhancing maritime safety in the Eastern Mediterranean Sea	Greece
2018	3rd Prize	Team Design, European BEST Engineering Competition · Description: Design and control of underwater vehicles	Greece
2018	2nd Prize	Team Design, European BEST Engineering Competition · Description: Design and control of hydraulic arms	Greece
2015	Certificate	Generic Engine Simulator course, Cosmos Nautical Training Centre · Description: Response to emergencies and rectification of system failures meeting the requirements of the STCW Code	Greece
2015	1st Prize	Group Project Competition in Electrosience, IMarEST · Topic: Lightning protection and grounding of marine vessels	Greece

Courses

2026	ProbAI	The 8 th Probabilistic AI School · Probabilistic Models; Variational Approximations; Deep Generative Models; Latent Variable Models; Normalizing Flows; etc.	Lithuania
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Information and Communication Technology Skills

Python; Matlab & Simulink	Training of data-driven AI controllers using DRL; Digital twins and system modeling; Autonomous operations and virtual environment simulations
High Performance Computing	More than 500,000 GPU/CPU core-hours of parallel computing for AI agents training
GeNIE; CARA; PTC Windchill	Risk analysis and risk modeling of autonomous systems using Bayesian belief networks, fault trees, and event trees
Qualisys	Motion capturing; System identification for the maneuvering model of ships
Towing Tank	Laboratory setup; Drag resistance measurements of ships
Rhinoceros 3D & Grasshopper; AVEVA Marine; AutoCAD	Design of ship's general arrangement plans and mechanical components; Parametric design of propellers

Scientific Productions

The comprehensive list of academic output is provided as finished works [1]-[14] and on-going works [15]-[18]:

- [01] Lee*, 2026. Towards risk-aware agents for autonomous vehicles – A collision probability approach using deep reinforcement learning. Engineering Applications of Artificial Intelligence. (*submitted*) NTNU
- [02] Lee*, Utne, Kim, 2026. Bridging the risk awareness gaps between the safety science and AI domains: A perspective with engineering applications. The 36th European Safety and Reliability Conference. NTNU
- [03] Grimstad, Jullian, Lee, Thieme, Ramos, Morozov, Utne, Mosleh, 2026. AI-Enabled Autonomy – The Trustworthiness Gap. The 7th International Workshop for Autonomous Systems Safety. NTNU
- [04] Grimstad, Jullian, Dugan, Lee, Ramos, Thieme, Morozov, Utne, Mosleh, 2025. Proceedings of the 6th International Workshop for Autonomous Systems Safety. The 6th International Workshop for Autonomous Systems Safety. NTNU
- [05] Lee*, Kim, 2025. Towards robust deep reinforcement learning agent for the path following of autonomous ships amid perception sensor noise. The 35th European Safety and Reliability Conference & The 33rd Society for Risk Analysis Europe Conference. NTNU
- [06] Lee, 2024. Robust decision making for the reactive collision avoidance of maritime autonomous surface ships. PhD thesis. UoS
- [07] Lee*, Theotokatos, Boulougouris, 2024. Robust decision-making for the reactive collision avoidance of autonomous ships against various perception sensor noise levels. Journal of Marine Science and Engineering. UoS
- [08] Lee, 2024. Robust reactive collision avoidance system for autonomous ships amidst limited sensory information and unclassified surface obstacles. TRA VISIONS 2024 Young Researcher Competition. UoS
- [09] Lee*, Theotokatos, Boulougouris, Bolbot, 2023. Risk-informed collision avoidance system design for maritime autonomous surface ships. Ocean Engineering. UoS
- [10] Lee, 2022. Trustworthy autonomous collision avoidance decision-making system for maritime autonomous surface ships. TRA VISIONS 2022 Young Researcher Competition. UoS
- [11] Jeon, Lee, Theotokatos, 2022. AUTOSHIP deliverable D7.2: autonomous ships: training framework for crew, operators and designers. AUTOSHIP. UoS
- [12] Lee, Bolbot, Theotokatos, Boulougouris, Vassalos, 2021. Fault tree analysis of the autonomous navigation for maritime autonomous surface ships. The 1st International Conference on the Stability and Safety of Ships and Ocean Vehicles. UoS
- [13] Lee, Liarokapis, Trachanas, Triantafyllou, Tzabiras, 2021. Experimental investigation of the effect of superhydrophobic coating and microbubbles lubrication on the resistance of a scaled ship. The 7th International Symposium on Ship Operations, Management and Economics. NTUA
- [14] Lee, 2020. Experimental investigation on the effect of superhydrophobic coating and microbubble lubrication on the resistance of a scaled ship. MEng thesis. NTUA
- [15] Jullian, Grimstad, Dugan, Thieme, Ramos, Lee, Morozov, Utne, Mosleh, 2026. Current Challenges and Recommendations for Stakeholders Involved with Autonomous Systems. (*draft version*) NTNU
- [16] Lee*, Theotokatos, Boulougouris, Utne, Kim, 2026. Risk-informed dynamic safety domain for the safe navigation of autonomous ships. (*draft version*) NTNU
- [17] Grimstad, Jullian, Lee, Thieme, Ramos, Morozov, Utne, Mosleh, 2026. The trustworthiness gap for autonomous systems. (*in preparation*) NTNU
- [18] Lee*, 2026. A combined supervised learning and reinforcement learning framework for the online classification of obstacles during collision avoidance. (*in preparation*) NTNU

References

Available upon request.